ICS 253 – Discrete Structures
Chapter 4: Functions

Question bank: 90 multiple choice + 10 written problems

Questions from ICS 253 old quizzes and homework (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

A function f that maps elements of a set X to elements of a set Y is a subset of $X \times Y$ such that:

(A) for every $y \in Y$ there is exactly one $x \in X$ with $(x, y) \in f$

(B) for every $x \in X$ there is at least one $y \in Y$ with $(x, y) \in f$

(C) for every $x \in X$ there is exactly one $y \in Y$ with $(x, y) \in f$

(D) for every $x \in X$ and every $y \in Y$, $(x, y) \in f$

Answer: (C). Every element of the domain is mapped to exactly one element of the target.

My answer

Q2. [new]

Let $X = \{w, x, y, z\}$, $A = \{a, b, c, d\}$ and $f : X \to A$ with $f = \{(w, a), (x, a), (y, d), (z, c)\}$. What is the range of f?

(A) $\{a, b, c, d\}$

(B) $\{a, c, d\}$

(C) $\{w, x, y, z\}$

(D) $\{a, d\}$

Answer: (B). The range is the set of target elements with at least one arrow coming in: a, c, d.

My answer

Q3. [new]

Let $X = \{w, x, y, z\}$ and $A = \{a, b, c, d\}$. Which of the following sets is a well-defined function from X to A?

(A) $\{(w, a), (x, a), (y, d), (z, c), (y, b)\}$

(B) $\{(w, a), (y, d), (z, c)\}$

(C) $\{(w, a), (x, b), (y, c), (d, z)\}$

(D) $\{(w, b), (x, b), (y, b), (z, b)\}$

Answer: (D). Each element of X appears exactly once. (A) maps y to two elements, (B) leaves x undefined, (C) contains $(d, z) \notin X \times A$.

My answer

Q4. [new]

In an arrow diagram for f with domain $\{a, b, c, d\}$ and target $\{w, x, y, z\}$, the arrows are $a \to x$, $b \to w$, $b \to y$, $c \to z$, $d \to x$. Is f a function?

(A) Yes, every element of the domain has an arrow leaving it

(B) No, because x has two arrows coming into it

(C) No, because b has two arrows leaving it

(D) No, because the range is not equal to the target

Answer: (C). In a function exactly one arrow leaves every domain element; b is mapped to both w and y.

My answer

Q5. [new]

In an arrow diagram for g with domain $\{a, b, c, d\}$ and target $\{w, x, y, z\}$, the arrows are $a \to z$, $b \to z$, $c \to w$, $d \to x$. Which statement is true?

(A) g is a function with range $\{w, x, z\}$

(B) g is not a function because z has two arrows coming in

(C) g is a function with range $\{w, x, y, z\}$

(D) g is not a function because y has no arrow coming in

Answer: (A). Exactly one arrow leaves each domain element, so g is a function; y is simply not in the range.

My answer

Q6. [new]

A function $f : \{a, b, c, d, e\} \to \{w, x, y, z\}$ has arrows $a \to z$, $b \to w$, $c \to y$, $d \to w$, $e \to z$. What is the range of f?

(A) $\{w, x, y, z\}$

(B) $\{a, b, c, d, e\}$

(C) $\{w, z\}$

(D) $\{w, y, z\}$

Answer: (D). Only w, y, z have arrows coming in; x does not.

My answer

Q7.

HW3 (Rosen 2.3 Ex 30a)

Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S) = \{f(s) : s \in S\}$ if $f(x) = 1$.

(A) $\emptyset$

(B) $\{1\}$

(C) $\{-1, 0, 1, 2, 4, 7\}$

(D) $\{0, 1\}$

Answer: (B). Every element of S is mapped to 1.

My answer

Q8.

[new]

Which of the following is a well-defined function from $\mathbb{R}$ to $\mathbb{R}$?

(A) $f(x) = \dfrac{1}{x}$

(B) $f(x) = \sqrt{x}$

(C) $f(x) = \sqrt{x^2 + 1}$

(D) $f(x) = \pm\sqrt{x^2 + 1}$

Answer: (C). (A) is undefined at 0, (B) is undefined for negative x, (D) maps each x to two values.

My answer

Q9.

[new]

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Which statement is correct?

(A) The target of f is $\{y \in \mathbb{R} : y \geq 0\}$ and the range is $\mathbb{R}$

(B) The target and the range of f are both $\mathbb{R}$

(C) The target of f is $\mathbb{R}$ and the range is $\{y \in \mathbb{R} : y \geq 0\}$

(D) The target and the range of f are both $\{y \in \mathbb{R} : y \geq 0\}$

Answer: (C). The target is given in the definition ($\mathbb{R}$); no number is mapped to a negative number, so the range is the non-negative reals.

My answer

Q10. [new]

Let $f : \{0,1\}^3 \to \{0,1\}^4$ be defined by $f(x) = x0$ (a 0 is added to the end of x). What is the range of f?

(A) All of $\{0,1\}^4$

(B) All 4-bit strings whose first bit is 0

(C) $\{0000\}$

(D) All 4-bit strings whose last bit is 0

Answer: (D). For example $f(101) = 1010$; every 4-bit string ending in 0 is hit, and none ending in 1.

My answer

Q11. [new]

Let $f : \{0,1\}^3 \to \mathbb{Z}$, where $f(x)$ is the number of 1s in x. What is the range of f?

(A) $\{0,1,2,3\}$

(B) $\{1,2,3\}$

(C) $\mathbb{Z}$

(D) $\{0,1,2,3,4,5,6,7\}$

Answer: (A). A 3-bit string contains between 0 and 3 ones, and every count occurs.

My answer

Q12. [new]

Let $f : \{0,1\}^5 \to \mathbb{Z}$, where $f(x)$ is the number of times "01" occurs in x (for example $f(01011) = 2$). What is the range of f?

(A) $\{1,2\}$

(B) $\{0,1,2\}$

(C) $\{0,1,2,3\}$

(D) $\{0,1,2,3,4\}$

Answer: (B). $f(11111) = 0$, $f(01111) = 1$, $f(01010) = 2$; three occurrences would need at least 6 bits.

My answer

Q13. [new]

Let $A = \{1,2,3\}$ and $f : A \times A \to \mathbb{Z}$ with $f(x,y) = xy$. What is the range of f?

(A) $\{1,2,3\}$

(B) $\{1,2,3,4,5,6,7,8,9\}$

(C) $\{1,2,3,4,6,9\}$

(D) $\{2, 3, 6\}$

Answer: (C). The products are $1, 2, 3, 2, 4, 6, 3, 6, 9$, i.e. $\{1, 2, 3, 4, 6, 9\}$.

My answer

Q14. [new]

In which of the following pairs is $f = g$?

(A) $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(x) = (x+1)^2$, $g(x) = x^2 + 1$

(B) $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$, $g(x) = |x|^2$

(C) $f, g : \mathbb{R} \to \mathbb{R}$, $f(x) = \lfloor x \rfloor$, $g(x) = \lceil x \rceil$

(D) $f, g : \mathbb{R} \to \mathbb{R}$, $f(x) = x$, $g(x) = \sqrt{x^2}$

Answer: (B). Same domain and target and $x^2 = |x|^2$ for all x. (A) differs at $x = 1$, (C) at $x = 0.5$, (D) at $x = -1$.

My answer

Q15. [new]

Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = \lceil x \rceil$ and $g : \mathbb{R} \to \mathbb{Z}$ with $g(x) = \lceil x \rceil$. Is $f = g$?

(A) Yes, because $f(x) = g(x)$ for every real x

(B) No, because f and g have different domains

(C) No, because $f(x) \neq g(x)$ for some real x

(D) No, because f and g have different targets

Answer: (D). Equal functions must have the same domain *and* the same target; here the targets are $\mathbb{R}$ and $\mathbb{Z}$.

My answer

Q16. [new]

Which of the following is correct?

(A) $\lfloor -1.3 \rfloor = -1$

(B) $\lceil -0.8 \rceil = -1$

(C) $\lfloor -1.3 \rfloor = -2$

(D) $\lceil 0.3 \rceil = 0$

Answer: (C). The floor slides down to the nearest integer: $-1.3 \to -2$. Also $\lceil -0.8 \rceil = 0$ and $\lceil 0.3 \rceil = 1$.

My answer

Q17. HW3 (Rosen 2.3 Ex 8f)

Find the value of $\lceil -2.99 \rceil$.

(A) -3

(B) -2

(C) 2

(D) 3

Answer: (B). The smallest integer ≥ -2.99 is -2.

My answer

Q18. HW3 (Rosen 2.3 Ex 8g)

Find the value of $\lfloor \frac{1}{2} + \lceil \frac{1}{2} \rceil \rfloor$.

(A) 0

(B) 2

(C) 1

(D) $\frac{3}{2}$

Answer: (C). $\lceil 1/2 \rceil = 1$, so $\lfloor 1/2 + 1 \rfloor = \lfloor 1.5 \rfloor = 1$.

My answer

Q19. HW3 (Rosen 2.3 Ex 8h)

Find the value of $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$.

(A) 2

(B) 1

(C) 3

(D) $\frac{3}{2}$

Answer: (A). $\lfloor 1/2 \rfloor = 0$, $\lceil 1/2 \rceil = 1$, so $\lceil 0 + 1 + 0.5 \rceil = \lceil 1.5 \rceil = 2$.

My answer

Q20. [new]

What is the value of $\lfloor -3.7 \rfloor + \lceil 2.1 \rceil$?

(A) -2

(B) 0

(C) -1

(D) 1

Answer: (C). $\lfloor -3.7 \rfloor = -4$ and $\lceil 2.1 \rceil = 3$, so the sum is -1.

My answer

Q21.

HW3 (Rosen 2.3 Ex 30c)

Let $S = \{-1, 0, 2, 4, 7\}$. Find $f(S) = \{f(s) : s \in S\}$ if $f(x) = \lceil x/5 \rceil$.

(A) $\{0, 1, 2\}$

(B) $\{-1, 0, 1, 2\}$

(C) $\{0, 1\}$

(D) $\{1, 2\}$

Answer: (A). $\lceil -0.2 \rceil = 0$, $\lceil 0 \rceil = 0$, $\lceil 0.4 \rceil = 1$, $\lceil 0.8 \rceil = 1$, $\lceil 1.4 \rceil = 2$.

My answer

Q22.

HW3 (Rosen 2.3 Ex 68g)

Let $f(x) = \left\lfloor 2\lceil x/2 \rceil + \frac{1}{2} \right\rfloor$ for real x with $-4 \le x \le 4$ (the function whose graph is drawn in the homework). What is the range of f on this interval?

(A) $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

(B) $\{-3, -1, 1, 3\}$

(C) $\{-4, -2, 0, 2, 4\}$

(D) $\{-2, -1, 0, 1, 2\}$

Answer: (C). $2\lceil x/2 \rceil$ is already an integer, so $f(x) = 2\lceil x/2 \rceil$, and $\lceil x/2 \rceil \in \{-2, -1, 0, 1, 2\}$ on $[-4, 4]$.

My answer

Q23.

[new]

A soccer team has 23 players. A pack of water contains six bottles. How many packs are needed so that every player gets one bottle?

(A) $\lfloor 23/6 \rfloor = 3$

(B) $\lceil 23/6 \rceil = 4$

(C) $\lceil 6/23 \rceil = 1$

(D) $\lfloor 23/6 \rfloor + 2 = 5$

Answer: (B). Three packs give only 18 bottles; we round $23/6 \approx 3.83$ up.

My answer

Q24. [new]

A farmer's hens lay 50 eggs. How many cartons can the farmer sell if each carton must contain exactly a dozen eggs?

(A) $\lceil 50/12 \rceil = 5$

(B) $\lfloor 12/50 \rfloor = 0$

(C) $50 - 12 = 38$

(D) $\lfloor 50/12 \rfloor = 4$

Answer: (D). Only full cartons can be sold, so we round $50/12 \approx 4.17$ down.

My answer

Q25. Quiz 4 Sec3 (Ver A/B)

How many positive integers less than 1000 are divisible by 7 but not by 11?

(A) $\lfloor \frac{1000}{7} \rfloor - \lfloor \frac{1000}{11} \rfloor$

(B) $\lfloor \frac{1000}{7} \rfloor$

(C) $\lfloor \frac{1000}{7} \rfloor - \left\lfloor \frac{1000}{7 \times 11} \right\rfloor$

(D) More than 142

Answer: (C). There are $\lfloor 1000/7 \rfloor = 142$ multiples of 7; remove the $\lfloor 1000/77 \rfloor = 12$ that are also multiples of 11, giving 130.

My answer

Q26. [new]

Which of the following is true for every real number x and every integer n?

(A) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

(B) $\lceil 2x \rceil = 2\lceil x \rceil$

(C) $\lfloor x \rfloor = \lceil x \rceil$

(D) $\lfloor nx \rfloor = n \lfloor x \rfloor$

Answer: (A). Adding an integer shifts x and the integer below it by the same amount. (B) fails at $x = 0.5$, (C) at $x = 0.5$, (D) at $n = 2$, $x = 0.5$.

My answer

Q27. **[new]**

Which statement about real numbers x and y is FALSE?

(A) $\lfloor -x \rfloor = -\lceil x \rceil$ for all x

(B) $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all x, y

(C) $\lfloor x \rfloor \leq x \leq \lceil x \rceil$ for all x

(D) $\lceil x \rceil = x$ if and only if x is an integer

Answer: (B). Counterexample: $x = y = 0.5$ gives $\lfloor 1 \rfloor = 1$ but $\lfloor 0.5 \rfloor + \lfloor 0.5 \rfloor = 0$.

My answer

Q28. **[new]**

If n is an odd integer, then $\lfloor n/2 \rfloor$ equals:

(A) $(n+1)/2$

(B) $n/2$

(C) $(n-1)/2$

(D) $\lceil n/2 \rceil$

Answer: (C). Write $n = 2k+1$; then $n/2 = k + 1/2$ and $\lfloor n/2 \rfloor = k = (n-1)/2$.

My answer

Q29. **[new]**

Let $f : \mathbb{R} \to \mathbb{Z}$ with $f(x) = \lfloor x + \frac{1}{2} \rfloor$. What is $f(-2.5)$?

(A) -3

(B) -2

(C) -2.5

(D) 2

Answer: (B). $\lfloor -2.5 + 0.5 \rfloor = \lfloor -2 \rfloor = -2$. ($f$ rounds to the nearest integer, rounding halves up.)

My answer

Q30. **[new]**

For which real numbers x is $\lfloor x \rfloor = 3$?

(A) $3 < x \leq 4$

(B) $2 < x \leq 3$

(C) $3 \leq x \leq 4$

(D) $3 \leq x < 4$

Answer: (D). $\lfloor x \rfloor = n$ exactly when $n \leq x < n + 1$.

My answer

Q31. [new]

A function $f : X \to Y$ is one-to-one (injective) if:

(A) for every $y \in Y$ there is an $x \in X$ with $f(x) = y$

(B) $f(x_1) = f(x_2)$ implies $x_1 = x_2$

(C) $x_1 = x_2$ implies $f(x_1) = f(x_2)$

(D) the range of f equals Y

Answer: (B). Equivalently, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$. (A) and (D) define onto; (C) holds for every function.

My answer

Q32. [new]

A function f assigns employees to offices. What does it mean if f is onto?

(A) No two employees share an office

(B) Every employee has an office

(C) No office is empty

(D) Each office has exactly one employee

Answer: (C). Onto means every element of the target (office) has some employee mapped to it. (A) is one-to-one; (D) is a bijection.

My answer

Q33. [new]

Let $f : \{w, x, y, z\} \to \{a, b, c, d, e\}$ with $f(w) = c$, $f(x) = b$, $f(y) = a$, $f(z) = c$. Then f is:

(A) one-to-one but not onto

(B) onto but not one-to-one

(C) a bijection

(D) neither one-to-one nor onto

Answer: (D). $f(w) = f(z) = c$, so not one-to-one; d and e have no preimage, so not onto.

My answer

Q34.

HW3 (Rosen 2.3 Ex 22a)

Is $f : \mathbb{R} \to \mathbb{R}$, $f(x) = -3x + 4$, a bijection?

(A) Yes; it is one-to-one and onto

(B) No; it is one-to-one but not onto

(C) No; it is onto but not one-to-one

(D) No; it is neither one-to-one nor onto

Answer: (A). $-3x_1 + 4 = -3x_2 + 4 \Rightarrow x_1 = x_2$, and for any y, $x = (4 - y)/3$ gives $f(x) = y$.

My answer

Q35.

HW3 (Rosen 2.3 Ex 22b)

Is $f : \mathbb{R} \to \mathbb{R}$, $f(x) = -3x^2 + 7$, a bijection?

(A) Yes

(B) No; it is one-to-one but not onto

(C) No; it is neither one-to-one nor onto

(D) No; it is onto but not one-to-one

Answer: (C). $f(1) = f(-1) = 4$, and $f(x) \leq 7$ for all x, so e.g. 8 is not in the range.

My answer

Q36.

HW3 (Rosen 2.3 Ex 22c)

Consider $f(x) = \dfrac{x+1}{x+2}$. Is f a bijection from $\mathbb{R}$ to $\mathbb{R}$?

(A) Yes

(B) No; it is one-to-one from $\mathbb{R}$ to $\mathbb{R}$ but not onto

(C) No; it is onto but not one-to-one

(D) No; it is not even a well-defined function from $\mathbb{R}$ to $\mathbb{R}$, since $f(-2)$ is undefined

Answer: (D). $f(-2)$ is undefined (and even on $\mathbb{R} - \{-2\}$ the value 1 is never taken). (old answer said "yes")

My answer

Q37. HW3 (Rosen 2.3 Ex 22d)

Is $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^5 + 1$, a bijection?

(A) No; it is not onto

(B) Yes; its inverse is $f^{-1}(y) = \sqrt[5]{y - 1}$

(C) No; it is not one-to-one

(D) Yes; its inverse is $f^{-1}(y) = \sqrt[5]{y} - 1$

Answer: (B). x^5 is strictly increasing and takes every real value; solving $y = x^5 + 1$ gives $x = \sqrt[5]{y - 1}$.

My answer

Q38. [new]

Let $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x$. Then f is:

(A) one-to-one but not onto

(B) onto but not one-to-one

(C) a bijection

(D) neither one-to-one nor onto

Answer: (A). $2x_1 = 2x_2 \Rightarrow x_1 = x_2$; odd integers such as 1 are not in the range.

My answer

Q39. [new]

Let $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = \lfloor x/2 \rfloor$. Then f is:

(A) one-to-one but not onto

(B) a bijection

(C) onto but not one-to-one

(D) neither one-to-one nor onto

Answer: (C). $f(0) = f(1) = 0$, so not one-to-one; for any y, $f(2y) = y$, so onto.

My answer

Q40. [new]

Which of the following functions $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto?

(A) $f(x, y) = 2x - 4y$

(B) $f(x, y) = x^2 + y^2$

(C) $f(x, y) = |x| + |y|$

(D) $f(x, y) = x + y$

Answer: (D). For any n, $f(n, 0) = n$. (A) gives only even numbers; (B) and (C) never give negatives.

My answer

Q41. [new]

Let $f : X \to Y$ with $|X| = 5$ and $|Y| = 7$. Which is possible?

(A) f is one-to-one

(B) f is onto

(C) f is a bijection

(D) None of these

Answer: (A). Onto requires $|X| \geq |Y|$ and a bijection requires $|X| = |Y|$; one-to-one only needs $|X| \leq |Y|$.

My answer

Q42. [new]

Let $f : \{0,1\}^3 \to \{0,1\}^2$, where $f(x)$ is obtained by dropping the first bit of x (e.g. $f(101) = 01$). Then f is:

(A) one-to-one but not onto

(B) onto but not one-to-one

(C) a bijection

(D) neither

Answer: (B). $f(001) = f(101) = 01$; every 2-bit string s equals $f(0s)$.

My answer

Q43. [new]

Let $f : \{0,1\}^3 \to \{0,1\}^3$, where $f(x)$ replaces the first bit of x by 1 (e.g. $f(010) = 110$). Then f is:

(A) neither one-to-one nor onto

(B) one-to-one but not onto

(C) onto but not one-to-one

(D) a bijection

Answer: (A). $f(010) = f(110)$, and 000 is not in the range. (For a finite set mapped to itself, one-to-one and onto go together.)

My answer

Q44. [new]

Let $f : \{0,1\}^3 \to \{0,1\}^4$, where $f(x)$ is obtained by adding a copy of the first bit to the end of x (e.g. $f(100) = 1001$). Then f is:

(A) onto but not one-to-one

(B) a bijection

(C) one-to-one but not onto

(D) neither

Answer: (C). The first three bits of $f(x)$ recover x, so one-to-one; 0001 is not in the range ($|\{0,1\}^3| < |\{0,1\}^4|$ anyway).

My answer

Q45. [new]

Let $A = \{1,2,3\}$ and $f : P(A) \to P(A)$ with $f(X) = X \cup \{1\}$. Then f is:

(A) a bijection

(B) one-to-one but not onto

(C) onto but not one-to-one

(D) neither one-to-one nor onto

Answer: (D). $f(\emptyset) = f(\{1\}) = \{1\}$, and $\{2\}$ is not in the range (every output contains 1).

My answer

Q46. [new]

Which function $f : \mathbb{Z} \to \mathbb{Z}^+$ is one-to-one but NOT onto?

(A) $f(x) = |x| + 1$

(B) $f(x) = 2x + 2$ if $x \geq 0$, and $f(x) = -2x + 1$ if $x < 0$

(C) $f(x) = 2x$ if $x > 0$, and $f(x) = -2x + 1$ if $x \leq 0$

(D) $f(x) = x^2 + 1$

Answer: (B). It gives $2, 4, 6, \ldots$ for $x \geq 0$ and $3, 5, 7, \ldots$ for $x < 0$: no repeats, but 1 is missed. (A) is onto not one-to-one, (C) is a bijection, (D) is neither.

My answer

Q47. [new]

Which statement is correct for a function $f : X \to Y$?

(A) f has an inverse if and only if f is one-to-one

(B) f has an inverse if and only if f is onto

(C) f has an inverse if and only if f is a bijection

(D) Every function has an inverse

Answer: (C). Reversing the arrows gives a function exactly when every y has exactly one incoming arrow.

My answer

Q48. [new]

$f : \{a, b, c, d\} \to \{v, w, x, y, z\}$ has arrows $a \to x$, $b \to v$, $c \to z$, $d \to w$. Does f have an inverse?

(A) Yes, because f is one-to-one

(B) No, because f is not one-to-one

(C) Yes, $f^{-1} = \{(x, a), (v, b), (z, c), (w, d)\}$

(D) No, because f is not onto: y would have no image under f^{-1}

Answer: (D). f is one-to-one but not onto, so reversing the arrows leaves y with no outgoing arrow.

My answer

Q49. [new]

Let $X = \{1, 2, 3\}$, $Y = \{7, 8, 9\}$ and $g : X \to Y$ with $g = \{(1, 9), (2, 7), (3, 8)\}$. What is $g^{-1}(8)$?

(A) 3

(B) 1

(C) 2

(D) g^{-1} is not defined

Answer: (A). g is a bijection and $g(3) = 8$, so $g^{-1}(8) = 3$.

My answer

Q50. [new]

$f : \{a, b, c, d, e\} \to \{m, n, p, q, r\}$ has arrows $a \to p$, $b \to m$, $c \to r$, $d \to n$, $e \to q$. What is $f^{-1}(m)$?

(A) a

(B) p

(C) b

(D) undefined

Answer: (C). f is a bijection and $f(b) = m$.

My answer

Q51. [new]

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x - 2$. What is $f^{-1}(y)$?

(A) $3y + 2$

(B) $\dfrac{y+2}{3}$

(C) $\dfrac{y-2}{3}$

(D) $\dfrac{1}{3y-2}$

Answer: (B). Solve $3x - 2 = y$: $x = (y + 2)/3$.

My answer

Q52. [new]

Let $f : \mathbb{R}^+ \to \mathbb{R}^+$, $f(x) = x^2$. Which is correct?

(A) f has no inverse because $f(-3) = f(3)$

(B) $f^{-1}(y) = y^2$

(C) $f^{-1}(y) = 1/y^2$

(D) f is a bijection and $f^{-1}(9) = 3$

Answer: (D). On the positive reals x^2 is one-to-one and onto, and $f^{-1}(y) = \sqrt{y}$.

My answer

Q53. [new]

Let $f : \{0,1\}^3 \to \{0,1\}^3$, where $f(x)$ moves the middle bit of x to the beginning (e.g. $f(011) = 101$). What is $f^{-1}(010)$?

(A) 001

(B) 010

(C) 100

(D) 110

Answer: (C). We need $x_1x_2x_3$ with $x_2x_1x_3 = 010$, so $x_2 = 0$, $x_1 = 1$, $x_3 = 0$: $x = 100$. Check: $f(100) = 010$.

My answer

Q54. [new]

Which of the following functions from $\{0,1\}^3$ to $\{0,1\}^3$ has a well-defined inverse?

(A) $f(x) = x$ with its bits reversed

(B) $f(x) = x$ with its first bit replaced by 1

(C) $f(x) = x$ with its first bit removed and a 1 added to the end

(D) $f(x) = x$ with its last bit replaced by a copy of its first bit

Answer: (A). Reversal is a bijection and is its own inverse. The others are not onto: (B) and (C) never output 000, (D) never outputs 001.

My answer

Q55. [new]

Let $A = \{1,2,3\}$ and $f : P(A) \to P(A)$ with $f(X) = A - X$. What is $f^{-1}(\{1\})$?

(A) $\{1\}$

(B) $\{2,3\}$

(C) $\emptyset$

(D) f^{-1} does not exist

Answer: (B). f is a bijection (it is its own inverse), and $f(\{2,3\}) = \{1\}$.

My answer

Q56. [new]

In the encryption example, a sender computes $y = f(x)$ and the receiver recovers x. Why must the encryption function f be a bijection?

(A) So that $f(x) = x$ for every message x

(B) So that the receiver can apply f^{-1} to y and obtain the unique original x

(C) So that the encrypted message is shorter than the original

(D) So that f is easy to guess

Answer: (B). Only a bijection has a well-defined inverse, which the receiver applies to decrypt.

My answer

Q57. [new]

Let $f : X \to Y$ and $g : Y \to Z$. The composition $g \circ f$:

(A) has domain Y and target Z, with $(g \circ f)(x) = f(g(x))$

(B) has domain X and target Z, with $(g \circ f)(x) = g(f(x))$

(C) has domain Z and target X, with $(g \circ f)(x) = g(f(x))$

(D) has domain X and target Y, with $(g \circ f)(x) = g(x)f(x)$

Answer: (B). f is applied first, then g.

My answer

Q58. [new]

Let $X = \{1, 2, 3\}$, $Y = \{a, b, c\}$, $Z = \{7, 10, 12\}$, $f : X \to Y$ with $f(1) = f(2) = a$, $f(3) = b$, and $g : Y \to Z$ with $g(a) = g(b) = 10$, $g(c) = 12$. What is the range of $g \circ f$?

(A) $\{10, 12\}$

(B) $\{7, 10, 12\}$

(C) $\{10\}$

(D) $\{a, b\}$

Answer: (C). Every element of X goes to a or b, and both go to 10.

My answer

Q59. HW3 (Rosen 2.3 Ex 36)

Let $f(x) = x^2 + 1$ and $g(x) = x + 2$ be functions from $\mathbb{R}$ to $\mathbb{R}$. What is $(f \circ g)(x)$?

(A) $x^2 + 3$

(B) $x^2 + 4x + 5$

(C) $x^2 + 2x + 3$

(D) $(x^2 + 1)(x + 2)$

Answer: (B). $f(g(x)) = (x + 2)^2 + 1 = x^2 + 4x + 5$.

My answer

Q60. HW3 (Rosen 2.3 Ex 36)

Let $f(x) = x^2 + 1$ and $g(x) = x + 2$ be functions from $\mathbb{R}$ to $\mathbb{R}$. What is $(g \circ f)(x)$?

(A) $x^2 + 4x + 5$

(B) $x^2 + 1$

(C) $(x + 2)^2$

(D) $x^2 + 3$

Answer: (D). $g(f(x)) = (x^2 + 1) + 2 = x^2 + 3$. Note $f \circ g \neq g \circ f$.

My answer

Q61. [new]

Let $f, g : \mathbb{Z} \to \mathbb{Z}$ with $f(x) = x^2$ and $g(x) = x + 1$. What is $(f \circ g)(-3)$?

(A) 4

(B) 10

(C) -8

(D) 9

Answer: (A). $g(-3) = -2$ and $f(-2) = 4$. (By contrast $(g \circ f)(-3) = 10$.)

My answer

Q62. [new]

$f : \{v, w, x, y, z\} \to \{a, b, c, d, e\}$ has $f(v) = b$, $f(w) = d$, $f(x) = e$, $f(y) = a$, $f(z) = b$, and $g : \{a, b, c, d, e\} \to \{1, 2, 3, 4, 5\}$ has $g(a) = 3$, $g(b) = 4$, $g(c) = 5$, $g(d) = 1$, $g(e) = 4$. What is the range of $g \circ f$?

(A) $\{1, 2, 3, 4, 5\}$

(B) $\{1, 3, 4, 5\}$

(C) $\{3, 4\}$

(D) $\{1, 3, 4\}$

Answer: (D). $g \circ f$: $v \mapsto 4$, $w \mapsto 1$, $x \mapsto 4$, $y \mapsto 3$, $z \mapsto 4$.

My answer

Q63. [new]

$f : \{a, b, c\} \to \{1, 2, 3, 4\}$ has $f(a) = 3$, $f(b) = f(c) = 2$, and $g : \{1, 2, 3, 4\} \to \{v, w, x, y\}$ has $g(1) = x$, $g(2) = w$, $g(3) = y$, $g(4) = v$. Which statement is true?

(A) $g \circ f$ is well-defined and $(g \circ f)(a) = y$

(B) $f \circ g$ is well-defined and $(f \circ g)(3) = y$

(C) $g \circ f$ is well-defined and is one-to-one

(D) $g \circ f$ is not well-defined

Answer: (A). The target of f is the domain of g, so $g \circ f$ exists; $g(f(a)) = g(3) = y$. It is not one-to-one since $b, c \mapsto w$. $f \circ g$ is not defined.

My answer

Q64. [new]

Let $f, h : \{0,1\}^3 \to \{0,1\}^3$, where $f(x)$ replaces the first bit of x by 1 and $h(x)$ reverses the bits of x. What is $(h \circ f)(010)$?

(A) 110

(B) 011

(C) 010

(D) 111

Answer: (B). $f(010) = 110$, then $h(110) = 011$. (In the other order, $(f \circ h)(010) = 110$.)

My answer

Q65. [new]

Let $f : X \to Y$ be a bijection. Which statement is true?

(A) $f^{-1} \circ f = I_Y$ and $f \circ f^{-1} = I_X$

(B) $f \circ f = I_X$

(C) $f^{-1} \circ f = I_X$ and $f \circ f^{-1} = I_Y$

(D) $f^{-1} \circ f = f$

Answer: (C). $f^{-1} \circ f$ starts and ends in X, and $f \circ f^{-1}$ starts and ends in Y.

My answer

Q66. [new]

For functions $f : A \to B$, $g : B \to C$, $h : C \to D$, which statement is always true?

(A) $g \circ f = f \circ g$

(B) $(h \circ g) \circ f = h \circ (g \circ f)$

(C) $h \circ g \circ f = f \circ g \circ h$

(D) $(h \circ g) \circ f$ has domain D

Answer: (B). Composition is associative (but not commutative).

My answer

Q67. [new]

Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x + 1$ and $g(x) = x^2$. What is $(g \circ f)(x)$?

(A) $2x^2 + 1$

(B) $4x^2 + 1$

(C) $4x^2 + 4x + 1$

(D) $2x^2 + 2x + 1$

Answer: (C). $g(f(x)) = (2x + 1)^2 = 4x^2 + 4x + 1$.

My answer

Q68. [new]

Let $f : X \to Y$ and $g : Y \to Z$. If $g \circ f$ is onto, which of the following must be true?

(A) f is onto

(B) g is onto

(C) f and g are both onto

(D) g is one-to-one

Answer: (B). Every $z \in Z$ equals $g(f(x))$ for some x, so $z = g(y)$ with $y = f(x)$. f need not be onto.

My answer

Q69. HW3 (Rosen 2.3 Ex 34)

If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one?

(A) No; e.g. $f(x) = (x + 1)/x$ and $g(x) = x^2 - 1$ is a counterexample

(B) No; g is one-to-one only if f is also onto

(C) Yes, but only when $f \circ g$ is also onto

(D) Yes; if $g(x_1) = g(x_2)$ then $f(g(x_1)) = f(g(x_2))$, so $x_1 = x_2$ because $f \circ g$ is one-to-one

Answer: (D). In fact $f \circ g$ one-to-one alone forces g one-to-one. The proposed counterexample fails: there $f \circ g$ is not one-to-one. (old answer said "no")

My answer

Q70. [new]

Let $f : X \to Y$ and $g : Y \to Z$ be one-to-one. Which is true about $g \circ f$?

(A) $g \circ f$ is one-to-one

(B) $g \circ f$ is onto

(C) $g \circ f$ need not be one-to-one

(D) $g \circ f$ is a bijection

Answer: (A). $g(f(x_1)) = g(f(x_2)) \Rightarrow f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

My answer

Q71. [new]

Two sets A and B have the same cardinality, $|A| = |B|$, if and only if:

(A) $A \subseteq B$ and $B \subseteq A$

(B) both are infinite

(C) there is a one-to-one function from A to B

(D) there is a one-to-one correspondence (bijection) from A to B

Answer: (D). This is Definition 1. A one-to-one function only gives $|A| \leq |B|$.

My answer

Q72. [new]

We write $|A| \leq |B|$ when:

(A) there is a one-to-one function from A to B

(B) there is an onto function from A to B

(C) A is finite and B is infinite

(D) $A \cap B = \emptyset$

Answer: (A). This is Definition 2.

My answer

Q73. [new]

A set is called countable if:

(A) it is infinite

(B) it is finite or has the same cardinality as $\mathbb{Z}^+$

(C) it is a subset of $\mathbb{R}$

(D) it has cardinality strictly less than $|\mathbb{Z}^+|$

Answer: (B). An infinite countable set has cardinality $\aleph_0$.

My answer

Q74. [new]

Which function is a one-to-one correspondence from $\mathbb{Z}^+$ to the set of odd positive integers?

(A) $f(n) = 2n$

(B) $f(n) = 2n + 1$

(C) $f(n) = 2n - 1$

(D) $f(n) = n^2$

Answer: (C). $2n - 1$ lists $1, 3, 5, \ldots$ with no repeats. $2n + 1$ misses 1.

My answer

Q75. Quiz 3 Sec3 (Ver A/B)

Consider the statements: (1) The positive integers less than $1{,}999{,}888{,}777$ are finite. (2) The integers that are multiples of 13 are countably infinite. (3) The real numbers between 0 and 0.2 are uncountable. (4) The real numbers between 10 and 100 are uncountable. Which are correct?

(A) Only 1 and 2

(B) Only 1 and 3

(C) All the statements are correct

(D) Only 2, 3 and 4

Answer: (C). A bounded set of positive integers is finite; multiples of 13 can be listed $0, 13, -13, 26, \ldots$; any interval of reals is uncountable.

My answer

Q76. Quiz 3 Sec3 (Ver A/B)

Consider: the set of all prime numbers; the set of all real numbers between 0 and 1; the set of all rational numbers; the set of all positive integers. Which of these sets are countably infinite?

(A) Only the positive integers

(B) Only the rationals and the positive integers

(C) The primes, the rationals and the positive integers

(D) All four sets

Answer: (C). The primes are an infinite subset of $\mathbb{Z}^+$, hence countably infinite; the reals in $(0, 1)$ are uncountable. (old key marked only the rationals and positive integers)

My answer

Q77. Quiz03 (Term 232)

Let $\mathbb{Z}$, $\mathbb{R}$, $\mathbb{Q}$ and $\overline{\mathbb{Q}}$ be the sets of integers, real, rational and irrational numbers. Which of the following is FALSE?

(A) $|\mathbb{Z}| = |\overline{\mathbb{Q}}|$

(B) $|\mathbb{R}| = |\overline{\mathbb{Q}}|$

(C) $|\mathbb{Z}| = |\mathbb{Q}|$

(D) $|\mathbb{Z}^+| = |\mathbb{Z}|$

Answer: (A). $\mathbb{Z}$ is countable but the irrationals are uncountable. (The original also listed $|\mathbb{Q}| = |\overline{\mathbb{Q}}|$, which is false too; it is replaced here so only one choice is false.)

My answer

Q78.

HW4 (Rosen 2.5 Ex 2a)

The set of integers greater than 10 is countably infinite. Which is a one-to-one correspondence from $\mathbb{Z}^+$ to this set?

(A) $f(n) = n + 10$

(B) $f(n) = 10n$

(C) $f(n) = n + 9$

(D) $f(n) = n - 10$

Answer: (A). $f(1) = 11, f(2) = 12, \ldots$; it is one-to-one and every integer $m > 10$ equals $f(m - 10)$.

My answer

Q79.

HW4 (Rosen 2.5 Ex 2b)

The set of odd negative integers is countably infinite. Which is a one-to-one correspondence from $\mathbb{Z}^+$ to this set?

(A) $f(n) = -2n$

(B) $f(n) = 1 - 2n$

(C) $f(n) = -2n - 1$

(D) $f(n) = -n$

Answer: (B). $f(1) = -1, f(2) = -3, f(3) = -5, \ldots$ ($-2n - 1$ misses -1).

My answer

Q80.

HW4 (Rosen 2.5 Ex 2e)

Let $A = \{2, 3\}$. The set $A \times \mathbb{Z}^+$ is:

(A) finite

(B) uncountable

(C) countably infinite, listed as $(2, 1), (3, 1), (2, 2), (3, 2), (2, 3), (3, 3), \ldots$

(D) countably infinite, but no list of it exists

Answer: (C). $f(n) = (2, (n+1)/2)$ for odd n and $f(n) = (3, n/2)$ for even n is a bijection from $\mathbb{Z}^+$.

My answer

Q81.

HW4 (Rosen 2.5 Ex 2f)

The set of integers that are multiples of 10 is countably infinite. Which sequence lists it (giving a one-to-one correspondence with $\mathbb{Z}^+$)?

(A) $10, 20, 30, 40, \ldots$

(B) $0, 10, 20, 30, \ldots$

(C) $-10, -20, -30, \ldots$

(D) $0, 10, -10, 20, -20, 30, -30, \ldots$

Answer: (D). Only this list contains every multiple of 10 (positive, negative and zero) exactly once.

My answer

Q82.

[new]

Which sequence corresponds to the one-to-one correspondence $f : \mathbb{Z}^+ \to \mathbb{Z}$, $f(n) = n/2$ if n is even and $f(n) = -(n-1)/2$ if n is odd, i.e. $f(1), f(2), f(3), \ldots$?

(A) $0, 1, -1, 2, -2, 3, -3, \ldots$

(B) $1, -1, 2, -2, 3, -3, \ldots$

(C) $0, -1, 1, -2, 2, \ldots$

(D) $1, 2, 3, 4, \ldots$

Answer: (A). $f(1) = 0$, $f(2) = 1$, $f(3) = -1$, $f(4) = 2$, $f(5) = -2, \ldots$

My answer

Q83.

[new]

Hilbert's Grand Hotel is fully occupied and one new guest arrives. How can the guest be accommodated without evicting anyone?

(A) It is impossible because all rooms are occupied

(B) Move the guest in room n to room $n+1$ and give room 1 to the new guest

(C) Move the guest in room n to room $2n$ and give room 2 to the new guest

(D) Give the new guest the last room

Answer: (B). Every current guest still has a room and room 1 becomes free. There is no last room.

My answer

Q84.

HW4 (Rosen 2.5 Ex 8)

A countably infinite number of new guests arrive at Hilbert's fully occupied Grand Hotel. Which plan gives every guest a room without evicting any current guest?

(A) Move the guest in room n to room $n+1$

(B) Move the guest in room n to room $2n$, then put the k-th new guest in room $2k-1$

(C) Move the guest in room n to room n^2, then put the new guests in rooms $2, 3, 5, 6, \ldots$ in order, which uses all remaining rooms

(D) It is impossible

Answer: (B). Current guests take the even rooms; the odd rooms $1, 3, 5, \ldots$ go to the new guests. (A) frees only one room.

My answer

Q85.

[new]

When the positive rationals p/q are listed by increasing $p+q$ (the zig-zag listing), the fraction $2/2$ is skipped. Why?

(A) Because $2/2$ is not a rational number

(B) Because the list only includes fractions with $p < q$

(C) Because $2/2 = 1$ has already been listed

(D) Because $p+q=4$ is even

Answer: (C). Each positive rational must appear exactly once, so repeats such as $2/2 = 1/1$ are left out.

My answer

Q86.

[new]

In Cantor's diagonal argument, suppose $r_1 = 0.23794\ldots$, $r_2 = 0.44590\ldots$, $r_3 = 0.09118\ldots$, $r_4 = 0.80553\ldots$, and the new number $r = 0.d_1d_2d_3d_4\ldots$ uses $d_i = 4$ if $d_{ii} \neq 4$ and $d_i = 5$ if $d_{ii} = 4$. The first four digits of r are:

(A) $0.4444\ldots$

(B) $0.5455\ldots$

(C) $0.2415\ldots$

(D) $0.4544\ldots$

Answer: (D). Diagonal digits are $2, 4, 1, 5$, giving $4, 5, 4, 4$. Then r differs from every r_i in the i-th place.

My answer

Q87. **[new]**

Which statement is correct?

(A) A subset of an uncountable set is uncountable

(B) A set with an uncountable subset is uncountable

(C) A subset of a countable set can be uncountable

(D) Every infinite set is countable

Answer: (B). If S were countable, every subset of S would be countable. (A) fails: $\mathbb{Z} \subseteq \mathbb{R}$.

My answer

Q88. HW4 (Rosen 2.5 Ex 10a)

Which pair of uncountable sets A, B has $A - B$ finite?

(A) $A = \mathbb{R}$, $B = \mathbb{R}^+$

(B) $A = \mathbb{R}$, $B = \mathbb{Q}$

(C) $A = \{x : 0 \leq x \leq 1\}$, $B = \{x : 0 < x < 1\}$

(D) $A = \mathbb{R}$, $B = \mathbb{R} - \mathbb{Z}$

Answer: (C). $A - B = \{0, 1\}$. (A) gives the non-positive reals, (B) the irrationals, (D) $\mathbb{Z}$.

My answer

Q89. HW4 (Rosen 2.5 Ex 10b)

Which pair of uncountable sets A, B has $A - B$ countably infinite?

(A) $A = \mathbb{R}$, $B = \mathbb{R} - \mathbb{Z}^+$

(B) $A = \mathbb{R}$, $B = \mathbb{R}^+$

(C) $A = \{x : 0 \leq x \leq 1\}$, $B = \{x : 0 < x < 1\}$

(D) $A = \mathbb{R}$, $B = \mathbb{R}$

Answer: (A). $A - B = \mathbb{Z}^+$, which is countably infinite. (This part was left blank in the homework.)

My answer

Q90. HW4 (Rosen 2.5 Ex 10c)

Which pair of uncountable sets A, B has $A - B$ uncountable?

(A) $A = \mathbb{R}$, $B = \mathbb{R}$

(B) $A = \{x : 0 \leq x \leq 1\}$, $B = \{x : 0 < x < 1\}$

(C) $A = \mathbb{R}$, $B = \mathbb{R} - \mathbb{Z}^+$

(D) $A = \mathbb{R}$, $B = \mathbb{R}^+$

Answer: (D). $A - B$ is the set of non-positive reals, which is uncountable.

My answer

Part B: Written problems

Q91. [new]

(a) Prove that for every real number x and every integer n, $\lfloor x+n \rfloor = \lfloor x \rfloor + n$.
(b) Prove or disprove: for all real numbers x and y, $\lceil x+y \rceil = \lceil x \rceil + \lceil y \rceil$.
(c) Prove that if n is an odd integer, then $\lfloor n/2 \rfloor = (n-1)/2$.

Answer:

(a) Let $m = \lfloor x \rfloor$. By definition m is an integer with $m \le x < m+1$. Adding the integer n gives

$$m + n \le x + n < (m+n) + 1,$$

and $m+n$ is an integer. The floor of a number is the unique integer k with $k \le \cdot < k+1$, so $\lfloor x+n \rfloor = m+n = \lfloor x \rfloor + n$. ■

(b) False. Take $x = y = 0.5$: $\lceil x+y \rceil = \lceil 1 \rceil = 1$, but $\lceil x \rceil + \lceil y \rceil = 1+1 = 2$.

(c) If n is odd then $n = 2k+1$ for some integer k, so $n/2 = k + \frac{1}{2}$. Since $k \le k + \frac{1}{2} < k+1$, we get $\lfloor n/2 \rfloor = k = (n-1)/2$. ■

My answer

Q92. [new]

Use the floor and ceiling functions to answer, showing the expression you use:
(a) 37 children each need five crayons. Crayons come in boxes of 24. How many boxes must be bought?
(b) A baker has 125 cookies and packs them in boxes of eight. How many full boxes can be sold?
(c) Compute $\lfloor 7.5 \rfloor + \lceil -7.5 \rceil$, $\lceil \lfloor 3.2 \rfloor + \frac{1}{2} \rceil$ and $\lfloor \lceil -1.5 \rceil / 2 \rfloor$.

Answer:

(a) $37 \cdot 5 = 185$ crayons are needed. Each box has 24, so we round up: $\lceil 185/24 \rceil = \lceil 7.708\ldots \rceil = 8$ boxes.

(b) Only full boxes can be sold, so we round down: $\lfloor 125/8 \rfloor = \lfloor 15.625 \rfloor = 15$ boxes.

(c) $\lfloor 7.5 \rfloor + \lceil -7.5 \rceil = 7 + (-7) = 0$.
$\lceil \lfloor 3.2 \rfloor + \frac{1}{2} \rceil = \lceil 3 + 0.5 \rceil = \lceil 3.5 \rceil = 4$.
$\lceil -1.5 \rceil = -1$, so $\lfloor -1/2 \rfloor = \lfloor -0.5 \rfloor = -1$.

My answer

Q93. [new]

Find a function $f : \mathbb{Z} \to \mathbb{Z}^+$ (domain all integers, target positive integers) that is: (a) neither one-to-one nor onto; (b) one-to-one but not onto; (c) onto but not one-to-one; (d) one-to-one and onto. Justify each answer.

Answer:

(a) $f(x) = x^2 + 1$. $f(1) = f(-1) = 2$, so not one-to-one; 3 is not in the range ($x^2 = 2$ has no integer solution), so not onto.

(b) $f(x) = 2x + 2$ for $x \geq 0$ and $f(x) = -2x + 1$ for $x < 0$. Non-negative x give the distinct even numbers $2, 4, 6, \ldots$; negative x give the distinct odd numbers $3, 5, 7, \ldots$; even and odd values never coincide, so f is one-to-one. The value 1 is never produced, so f is not onto.

(c) $f(x) = |x| + 1$. $f(1) = f(-1) = 2$, so not one-to-one; for any $y \in \mathbb{Z}^+$, $f(y - 1) = y$, so onto.

(d) $f(x) = 2x$ for $x > 0$ and $f(x) = -2x + 1$ for $x \leq 0$. Positive x give every even positive integer exactly once; $x = 0, -1, -2, \ldots$ give $1, 3, 5, \ldots$, every odd positive integer exactly once. Hence f is one-to-one and onto.

My answer

Q94. Quiz 3 Adel

Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$. Prove or disprove: "If f and g are both bijections, then $f + g$, defined as $(f + g)(x) = f(x) + g(x)$, is also a bijection."

Answer:

The statement is false. Counterexample: let $f(x) = x$ and $g(x) = -x$.

- f is a bijection: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, and $f(y) = y$ for every y.
- g is a bijection: $-x_1 = -x_2 \Rightarrow x_1 = x_2$, and $g(-y) = y$ for every y.

But $(f + g)(x) = x + (-x) = 0$ for all x. Then $(f + g)(0) = (f + g)(1)$, so $f + g$ is not one-to-one, and 1 is not in its range, so it is not onto. Hence $f + g$ is not a bijection.

My answer

Q95. Quiz 3 Adel

Let $g : A \to B$ and $f : B \to C$. Show that if both f and g are onto, then $f \circ g$ is onto.

Answer:

Let $c \in C$ be arbitrary. Since f is onto, there is $b \in B$ with $f(b) = c$. Since g is onto, there is $a \in A$ with $g(a) = b$. Then

$$(f \circ g)(a) = f(g(a)) = f(b) = c.$$

Since c was arbitrary, every element of C is in the range of $f \circ g$, so $f \circ g : A \to C$ is onto. ■

My answer

Q96. [new]

Let $f : X \to Y$ and $g : Y \to Z$.
(a) Prove that if $g \circ f$ is one-to-one, then f is one-to-one.
(b) Show by an example that $g \circ f$ can be one-to-one even though g is not one-to-one.

Answer:

(a) Suppose $f(x_1) = f(x_2)$. Applying g to both sides gives $g(f(x_1)) = g(f(x_2))$, that is $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since $g \circ f$ is one-to-one, $x_1 = x_2$. Hence f is one-to-one. ■

(b) Let $X = \{1\}$, $Y = \{a, b\}$, $Z = \{7\}$, with $f(1) = a$ and $g(a) = g(b) = 7$. Then g is not one-to-one ($g(a) = g(b)$), but $g \circ f$ has a one-element domain, so it is one-to-one.

My answer

Q97. [new]

Let $f : \mathbb{R} - \{2\} \to \mathbb{R} - \{1\}$ be defined by $f(x) = \dfrac{x+1}{x-2}$.
(a) Show that f is well-defined (that is, $f(x) \neq 1$ for all x in the domain).
(b) Prove that f is one-to-one and onto.
(c) Find f^{-1} and verify that $(f \circ f^{-1})(y) = y$.

Answer:

(a) If $\frac{x+1}{x-2} = 1$ then $x + 1 = x - 2$, i.e. $1 = -2$, impossible. So every output lies in $\mathbb{R} - \{1\}$, and the expression is defined because $x \neq 2$.

(b) One-to-one: suppose $\frac{x_1+1}{x_1-2} = \frac{x_2+1}{x_2-2}$. Cross-multiplying,

$$\begin{aligned}(x_1 + 1)(x_2 - 2) &= (x_2 + 1)(x_1 - 2)\\ x_1x_2 - 2x_1 + x_2 - 2 &= x_1x_2 - 2x_2 + x_1 - 2\\ -3x_1 &= -3x_2 \;\Rightarrow\; x_1 = x_2.\end{aligned}$$

Onto: let $y \neq 1$. Solve $\frac{x+1}{x-2} = y$: $x + 1 = xy - 2y$, so $x(1 - y) = -2y - 1$ and $x = \frac{2y+1}{y-1}$. This is defined since $y \neq 1$, and $x \neq 2$ because $\frac{2y+1}{y-1} = 2$ would give $2y + 1 = 2y - 2$, impossible. Then $f(x) = y$, so f is onto. Thus f is a bijection.

(c) From the computation above, $f^{-1} : \mathbb{R} - \{1\} \to \mathbb{R} - \{2\}$, $f^{-1}(y) = \dfrac{2y+1}{y-1}$. Check:

$$f(f^{-1}(y)) = \frac{\frac{2y+1}{y-1} + 1}{\frac{2y+1}{y-1} - 2} = \frac{2y + 1 + y - 1}{2y + 1 - 2y + 2} = \frac{3y}{3} = y.$$

My answer

Q98. **[new]**

Let $A = \{1, 2, 3\}$ and $f : P(A) \to P(A)$ be defined by $f(X) = A - X$.
(a) Compute $f(\emptyset)$, $f(\{1\})$ and $f(\{1, 3\})$.
(b) Prove that $f \circ f = I_{P(A)}$.
(c) Use (b) to conclude that f is a bijection and find f^{-1}.

Answer:

(a) $f(\emptyset) = \{1, 2, 3\}$, $f(\{1\}) = \{2, 3\}$, $f(\{1, 3\}) = \{2\}$.

(b) For any $X \subseteq A$, $(f \circ f)(X) = A - (A - X)$. An element $a \in A$ lies in $A - (A - X)$ iff $a \notin A - X$ iff $a \in X$. Hence $A - (A - X) = X$, so $(f \circ f)(X) = X$ for every X, i.e. $f \circ f = I_{P(A)}$. ■

(c) One-to-one: if $f(X_1) = f(X_2)$ then $X_1 = f(f(X_1)) = f(f(X_2)) = X_2$. Onto: any $Y \in P(A)$ equals $f(f(Y))$, so Y is the image of $f(Y)$. Thus f is a bijection, and since $f \circ f$ is the identity, $f^{-1} = f$, i.e. $f^{-1}(Y) = A - Y$.

My answer

Q99. **[new]**

Prove that the set $\mathbb{Z}$ of all integers is countable by showing that

$$f : \mathbb{Z}^+ \to \mathbb{Z}, \qquad f(n) = \begin{cases} n/2 & n \text{ even} \\ -(n-1)/2 & n \text{ odd} \end{cases}$$

is a one-to-one correspondence.

Answer:

Values: $f(1) = 0$, $f(2) = 1$, $f(3) = -1$, $f(4) = 2$, $f(5) = -2, \ldots$. Note that $f(n) > 0$ when n is even and $f(n) \leq 0$ when n is odd.

One-to-one: suppose $f(n_1) = f(n_2)$. If the common value is positive, both n_1, n_2 are even and $n_1/2 = n_2/2$, so $n_1 = n_2$. If it is ≤ 0, both are odd and $-(n_1 - 1)/2 = -(n_2 - 1)/2$, so $n_1 = n_2$.

Onto: let $m \in \mathbb{Z}$. If $m > 0$, then $n = 2m$ is an even positive integer with $f(n) = m$. If $m \leq 0$, then $n = 1 - 2m$ is an odd positive integer with $f(n) = -(1 - 2m - 1)/2 = m$.

Hence f is a bijection, $|\mathbb{Z}| = |\mathbb{Z}^+|$, and $\mathbb{Z}$ is countable (cardinality $\aleph_0$). ■

My answer

Q100. **[new]**

Let S be the set of all infinite sequences of bits $b_1 b_2 b_3 \ldots$ (each $b_i \in \{0, 1\}$). Use a Cantor diagonal argument to prove that S is uncountable.

Answer:

Suppose, for contradiction, that S is countable. S is infinite (e.g. it contains $10000\ldots$, $01000\ldots$, $00100\ldots$, $\ldots$), so its elements can be listed as $s_1, s_2, s_3, \ldots$, where

$$s_i = b_{i1} b_{i2} b_{i3} \ldots$$

Define a new sequence $t = t_1 t_2 t_3 \ldots$ by $t_i = 1 - b_{ii}$ (flip the i-th bit of the i-th sequence). Then $t \in S$. For every i, t differs from s_i in position i, since $t_i \neq b_{ii}$. Hence $t \neq s_i$ for all i, so t is not in the list. This contradicts the assumption that the list contains every element of S. Therefore S is uncountable. ■

My answer